Nicole Zagson

(201) 694-1960 ♦ gerzon.n@northeastern.edu ♦ linkedin.com/in/nicole-zagson/ ♦ nicolegerzon813.github.io

EDUCATION

Northeastern University, Cybersecurity and Privacy Lab, Boston, MA **Sept. 2024 - present**
PhD Candidate in Cybersecurity *GPA: 4.00/4.00*
Advised by Dr. Alan Mislove and Dr. Piotr Sapieżyński

Northeastern University, Khoury College of Computer Science, Boston, MA **Sept. 2019 - May 2024**
Candidate for Bachelor of Science in Cybersecurity *GPA: 3.92/4.00*

ACADEMIC RESEARCH

Spotify Recommendations (in progress) **July 2025 - present**
- Investigating Spotify's podcast recommendation algorithm to uncover the underlying structure of recommendations and how user activity affects episode and show recommendations.

Automatic Transmission: Study of Data Privacy in the Connected Vehicle Ecosystem **Jan. 2024 - Oct. 2026**
To be published at AMC IMC 2026
- Conducted a comprehensive analysis of data generated by connected vehicles and companion mobile apps
- Reverse-engineered mobile and in-vehicle apps to determine how personal/vehicle data is collected & processed
- Engaged in comprehensive disclosure process to disclose and better understand industry challenges in mitigating privacy risk

Quantifying Sandwiching MEV on Jito: Measurement of Solana's Leading Validator Client **Jan. - Oct. 2025**
Published at AMC IMC 2025
- Reverse engineered hidden APIs to collect 4 months worth of otherwise opaque Jito-Solana transaction data, identifying over 500k Sandwiching MEV attacks, over $7.7 million in cumulative user losses.
- Found evidence of

When For You Isn't "For You": Measuring User Agency in TikTok's Algorithmic Feed **Oct. 2022 - May 2026**
Published at ICWSM 2026
- Developed methods for TikTok algorithmic audits, distinguishing when personalization is taking effect, and how to trigger algorithm rabbit holes

Track You: A Deep Dive into Safety Alerts for Apple AirTags **Mar. 2022 - July 2023**
Published at PETS 2023
- Created methods to analyze the underlying structure and characteristics of Apple's FindMy network infrastructure
- Developed a process for creating stealthy, counterfeit AirTags and linking them to real keys, demonstrating flaws in Apple's data storage and security protocols

Measurement and Analysis of Implied Identity in Ad Delivery Optimization **Jan. 2021 - Oct. 2022**
Awarded Distinguished Long Paper at AMC Internet Measurement Conference 2022
- Discovered ad delivery is skewed based on demographics, where images of white people are delivered less to Black people, supporting a need for increased diversity in advertising

WORK EXPERIENCE

Texas Office of the Attorney General, Austin, TX
Cybersecurity Intern, Child Support Division **Aug. 2023 - Aug.2024**
- Facilitated compliance audits for IRS, CJIS, and various state-wide regulatory requirements
- Validated implementation and ensured documentation of NIST, FedRAMP, and TexRAMP security controls and policies across IT organization and third party vendors
- Scanned for and analyzed risk of open vulnerabilities across division infrastructure and state-wide offices for patch management purposes and presenting security information in weekly advisory board meetings

.